A DIRECTED COLIMIT OF FORMAL ALGEBRAIC SPACES GLUED BY CLOSED IMMERSIONS IS A FORMAL ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's own wording 'we see conclude that'; it lies inside the counted span and is reproduced verbatim, as the delivered text is the author's.

Editorial note. The proof displays one cartesian square; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The definition of a formal algebraic space through coverings by affine formal algebraic spaces. Same chapter, Definition definition-formal-algebraic-space, invoked at proof lines 7913-7915. Fixes what has to be produced: a covering of the colimit by affine formal algebraic spaces.
- The identification of the affine etale site with the category of representable etale affine formal algebraic spaces over the space. Same chapter, Lemma lemma-affine-identify-affine-etale, invoked at proof lines 7923-7927. Produces the etale morphisms over each member of the system corresponding to the chosen covering, and shows they are compatible.
- A directed colimit of affine formal algebraic spaces joined by closed immersions inducing isomorphisms on reductions is an affine formal algebraic space. Same chapter, Lemma lemma-colimit-affine-formal, invoked at proof lines 7949-7951. Gives that each piece of the constructed covering is affine.

Source reference. Original lemma 0DEE: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7480–7480. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GVM: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7868–7868. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Suppose given a directed set Λ and a system of formal algebraic spaces $(X_\lambda, f_{\lambda\mu})$ over Λ where each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a closed immersion inducing an isomorphism $X_{\lambda,red} \to X_{\mu,red}$. Then $X = \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$ is a formal algebraic space over S.*

Proof. Since we take the colimit in the category of fppf sheaves, we see that X is a sheaf. Choose and fix $\lambda \in \Lambda$. Choose a covering $\{X_{i,\lambda} \to X_\lambda\}$ as in Definition 0.1. In particular, we see that $\{X_{i,\lambda,red} \to X_{\lambda,red}\}$ is an étale covering by affine schemes. For each $\mu \geq \lambda$ there exists a cartesian diagram

$$\begin{array}{ccc} X_{i,\lambda} & \longrightarrow & X_{i,\mu} \\ \downarrow & & \downarrow \\ X_\lambda & \longrightarrow & X_\mu \end{array}$$

with étale vertical arrows. Namely, the étale morphism $X_{i,\lambda,red} \to X_{\lambda,red} = X_{\mu,red}$ corresponds to an étale morphism $X_{i,\mu} \to X_\mu$ of formal algebraic spaces with $X_{i,\mu}$ an affine formal algebraic space, see Lemma 0DEE. The same lemma implies the base change of $X_{i,\mu}$ to X_λ agrees with $X_{i,\lambda}$. It also follows that $X_{i,\mu} = X_\mu \times_{X_{\mu'}} X_{i,\mu'}$ for $\mu' \geq \mu \geq \lambda$. Set $X_i = \operatorname{colim} X_{i,\mu}$. Then $X_{i,\mu} = X_i \times_X X_\mu$ (as functors). Since any morphism $T \to X = \operatorname{colim} X_\mu$ from an affine (or quasi-compact) scheme T maps into X_μ for some μ, we see conclude that $\operatorname{colim} X_{i,\mu} \to \operatorname{colim} X_\mu$ is étale. Thus, if we can show that $\operatorname{colim} X_{i,\mu}$ is an affine formal algebraic space, then the lemma holds. Note that the morphisms $X_{i,\mu} \to X_{i,\mu'}$ are closed immersions as a base change of the closed immersion $X_\mu \to X_{\mu'}$. Finally, the morphism $X_{i,\mu,red} \to X_{i,\mu',red}$ is an isomorphism as $X_{\mu,red} \to X_{\mu',red}$ is an isomorphism. Hence this reduces us to the case discussed in Lemma 0GVM. □